SYNTOMICITY OF THE THIRD MORPHISM IN A SURJECTIVE SYNTOMIC COVER OVER A SYNTOMIC MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-syntomic-permanence is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 6057-6077 and proof span 6079-6127 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The statement carries a footnote pointing to a Descent chapter lemma for a stronger variant; the footnote is part of the counted statement span and is delivered verbatim, with the cited label handled as an external navigation pointer.

Editorial note. 7 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- Flatness permanence for the diagram, which the author cites to get flatness of q from the flatness of the syntomic morphisms in the square. Cited at proof line 3079. Reduces the claim that q is syntomic to the claim that its fibres are local complete intersections.
- The criterion that a flat morphism of finite presentation is syntomic exactly when its fibres are local complete intersections, together with the characterisation of local standard syntomic ring maps. Cited at proof lines 3080-3081 and again at 3120-3122 and 3124-3125. Forms the bridge between the geometric conclusion and the local algebra statement produced at the end of the argument.
- Base change of syntomic and of locally finite type morphisms, used to reduce the base to the spectrum of the residue field of a point. Cited at proof lines 3084-3088. Lets the proof work with S the spectrum of a field without loss of generality.
- The algebra lemma on the initial case of local complete intersection permanence, whose hypotheses are checked explicitly in the counted proof. Cited

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at proof lines 3111-3123. Supplies the conclusion that the local ring of Y at the chosen point is a complete intersection ring.

Source reference. Original reference algebra-lemma-lci-permanence-initial: sources/stacks-morphisms/pinned-original-source.tex, source lines 6115–6115. Same-chapter reference used in the retained author text at source line(s) 6115, 6123. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original reference descent-lemma-syntomic-permanence: sources/stacks-morphisms/pinned-original-source.tex, source lines 6074–6074. Same-chapter reference used in the retained author text at source line(s) 6074. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma lemma-base-change-finite-type: sources/stacks-morphisms/pinned-original-source.tex, source lines 2732–2732. Same-chapter reference used in the retained author text at source line(s) 2956, 3695, 3779, 3950, 4313, 6088, 9750, 9928. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma lemma-base-change-syntomic: sources/stacks-morphisms/pinned-original-source.tex, source lines 5839–5839. Same-chapter reference used in the retained author text at source line(s) 6085. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma lemma-flat-permanence: sources/stacks-morphisms/pinned-original-source.tex, source lines 4881–4881. Same-chapter reference used in the retained author text at source line(s) 6080, 7396, 7512, 8452. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma lemma-syntomic-flat-fibres: sources/stacks-morphisms/pinned-original-source.tex, source lines 5954–5954. Same-chapter reference used in the retained author text at source line(s) 6082. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma lemma-syntomic-locally-standard-syntomic: sources/stacks-morphisms/pinned-original-source.tex, source lines 5932–5932. Same-chapter reference used in the retained author text at source line(s) 5964, 6002, 6036, 6122, 6125. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let*

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & {\scriptstyle p}\searrow \quad \swarrow{\scriptstyle q} & \\ & S & \end{array}$$

be a commutative diagram of morphisms of schemes. Assume that

(1) *f is surjective and syntomic,*
(2) *p is syntomic, and*
(3) *q is locally of finite presentation*[1]*.*

Then q is syntomic.

Proof. By Lemma lemma-flat-permanence we see that q is flat. Hence it suffices to show that the fibres of $Y \to S$ are local complete intersections, see Lemma lemma-syntomic-flat-fibres. Let $s \in S$. Consider the morphism $X_s \to Y_s$. This is a base change of the morphism $X \to Y$ and hence surjective, and syntomic (Lemma lemma-base-change-syntomic). For the same reason X_s is syntomic over $\kappa(s)$. Moreover, Y_s is locally of finite type over $\kappa(s)$ (Lemma lemma-base-change-finite-type). In this way we reduce to the case where S is the spectrum of a field.

Assume $S = \mathrm{Spec}(k)$. Let $y \in Y$. Choose an affine open $\mathrm{Spec}(A) \subset Y$ neighbourhood of y. Let $\mathrm{Spec}(B) \subset X$ be an affine open such that $f(\mathrm{Spec}(B)) \subset \mathrm{Spec}(A)$, containing a point $x \in X$ such that $f(x) = y$. Choose a surjection $k[x_1, \ldots, x_n] \to A$ with kernel I. Choose a surjection $A[y_1, \ldots, y_m] \to B$, which gives rise in turn to a surjection $k[x_i, y_j] \to B$ with kernel J. Let $\mathfrak{q} \subset k[x_i, y_j]$ be the prime corresponding to $y \in \mathrm{Spec}(B)$ and let $\mathfrak{p} \subset k[x_i]$ the prime corresponding to $x \in \mathrm{Spec}(A)$. Since x maps to y we have $\mathfrak{p} = \mathfrak{q} \cap k[x_i]$. Consider the following commutative diagram of local rings:

$$\begin{array}{ccccc} \mathcal{O}_{X,x} & = & B_{\mathfrak{q}} & \leftarrow & k[x_1, \ldots, x_n, y_1, \ldots, y_m]_{\mathfrak{q}} \\ \uparrow & & \uparrow & & \uparrow \\ \mathcal{O}_{Y,y} & = & A_{\mathfrak{p}} & \leftarrow & k[x_1, \ldots, x_n]_{\mathfrak{p}} \end{array}$$

We claim that the hypotheses of Algebra, Lemma algebra-lemma-lci-permanence-initial are satisfied. Conditions (1) and (2) are trivial. Condition (4) follows as $X \to Y$ is flat. Condition (3) follows as the rings $\mathcal{O}_{Y,y}$ and $\mathcal{O}_{X_y,x} = \mathcal{O}_{X,x}/\mathfrak{m}_y\mathcal{O}_{X,x}$ are complete intersection rings by our assumptions that f and p are syntomic, see Lemma lemma-syntomic-locally-standard-syntomic. The output of Algebra, Lemma algebra-lemma-lci-permanence-initial is exactly that $\mathcal{O}_{Y,y}$ is a complete intersection ring! Hence by Lemma lemma-syntomic-locally-standard-syntomic again we see that Y is syntomic over k at y as desired. □

[1] In fact, if f is surjective, flat, and locally of finite presentation and p is syntomic, then both q and f are syntomic, see Descent, Lemma descent-lemma-syntomic-permanence.